

★ SI BASE + DERIVED UNITS ★

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► Meaning in one line

7 base , key derived , symbols.

► Key points

Point	Example
Base 7: m/kg/s/A/K/mol/cd + symbols.	-
Derived: N/J/W/V/ohm + formula link.	N = kg m per s ² .

★ IMPORTANT FOR EXAM

Prefixes + notation

⚠ TRAP - read twice

Name-only + 'unit asked' + symbol-miss + formula-miss

✓ **Try yourself:** Unit + symbol + formula + trio? **Answer:** What + symbol + link.

★ PREFIXES + SCIENTIFIC NOTATION ★

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► Meaning in one line

kilo/milli/micro/nano , powers , order.

► Key points

Point	Example
kilo/milli/micro/nano/pico + power.	-
Notation + order + conversion.	3.2 x 10 to power 5.

★ IMPORTANT FOR EXAM

Scalar/vector + distance

⚠ TRAP - read twice

Prefix-only + 'convert' + power-miss + direction-miss

✓ **Try yourself:** Prefix + power + direction + conversion? **Answer:** Value + power + shift.

★ SCALAR + VECTOR + DISTANCE ★

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► Meaning in one line

Magnitude-only vs magnitude, direction , path vs straight.

► Key points

Point	Example
Scalar: speed/distance/work/energy + magnitude only.	-
Vector: velocity/displacement/force + direction.	displacement zero on return.

★ IMPORTANT FOR EXAM

Speed/velocity + acceleration

⚠ TRAP - read twice

Magnitude-only + 'distance asked' + direction-miss + path-miss

✓ **Try yourself:** Scalar + vector + path + direction + sort? **Answer:** Magnitude + direction.

★ SPEED + VELOCITY + ACCELERATION ★

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► Meaning in one line

Rate , direction , change-rate , units.

► Key points

Point	Example
Speed scalar + velocity vector + units m per s.	-
Acceleration = change in velocity over time + m per s ² .	free fall g.

★ IMPORTANT FOR EXAM

Graphs + equations

⚠ TRAP - read twice

Value-only + 'fast' + direction-miss + unit-miss

✓ **Try yourself:** Rate + direction + unit + change? **Answer:** v + a + units.

★ MOTION GRAPHS (DIST + VEL TIME) ★

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► Meaning in one line

Slope , area , shape.

► Key points

Point	Example
Distance-time: slope = speed + flat = rest.	-
Velocity-time: slope = acceleration + area = displacement.	horizontal = uniform.

★ IMPORTANT FOR EXAM

Equations + force

⚠ TRAP - read twice

Shape-only + 'graph asked' + slope-miss + area-miss

✓ **Try yourself:** Slope + area + shape + graph + read? **Answer:** Slope + area.

★ MOTION EQUATIONS + FORCE ★

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► Meaning in one line

$v = u$, at , $s = ut$, $\frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $F = ma$.

► Key points

Point	Example
Motion: $v = u + at$ + $s = ut + \frac{1}{2}at^2$ + $v^2 = u^2 + 2as$.	-
Force: $F = ma$ + newton + net + direction.	double m = double F .

★ IMPORTANT FOR EXAM

Newton laws + momentum

⚠ TRAP - read twice

Formula-only + 'plug values' + unit-miss + net-miss

✓ **Try yourself:** Formula + unit + net + solve? **Answer:** Pick + plug + unit.

★ NEWTON 1 + NEWTON 2 ★

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► Meaning in one line

Inertia , $F = ma$, examples.

► Key points

Point	Example
First: inertia + rest/motion continues + unbalanced changes.	seat belt.
Second: $F = m a$ + net + direction.	heavier needs bigger F .

★ IMPORTANT FOR EXAM

Third law + momentum

⚠ TRAP - read twice

Statement-only + 'law asked' + example-miss + net-miss

✓ **Try yourself:** Law + example + net + inertia? **Answer:** State + example.

★ NEWTON 3 + MOMENTUM ★

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► Meaning in one line

Action-reaction , $p = mv$, impulse.

► Key points

Point	Example
Third: equal opposite + different bodies.	rocket push.
Momentum $p = m v$ + impulse = $F t$ + conserved.	recoil.

★ IMPORTANT FOR EXAM

Work + power

⚠ TRAP - read twice

Single-body + 'reaction' + pair-miss + direction-miss

✓ **Try yourself:** Pair + product + direction + conserved? **Answer:** Action + p.

★ WORK + POWER ★

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► Meaning in one line

$W = F d$, $P = W \text{ over } t$, units.

► Key points

Point	Example
Work = $F \times d$ along force + joule.	zero when perpendicular.
Power = work over time + watt.	1 HP link.

★ IMPORTANT FOR EXAM

KE + PE + conservation

⚠ TRAP - read twice

Value-only + 'work asked' + angle-miss + unit-miss

✓ **Try yourself:** $F d$ + rate + unit + angle? **Answer:** $W + P$ + units.

★ KINETIC + POTENTIAL ENERGY ★

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► Meaning in one line

KE = half $m v^2$, PE = $m g h$, units.

► Key points

Point	Example
KE = half $m v^2$ + speed matters squared.	double $v = 4 \times$ KE.
PE = $m g h$ + height.	raised water + wound spring.

★ IMPORTANT FOR EXAM

Conservation + machines

⚠ TRAP - read twice

Formula-only + 'energy asked' + squared-miss + unit-miss

✓ **Try yourself:** Motion + height + squared + unit? **Answer:** KE + PE.

★ CONSERVATION + MACHINES ★

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► Meaning in one line

Total constant , $MA = \text{load over effort}$.

► Key points

Point	Example
Conservation: total same + KE/PE swap.	pendulum + falling body.
Machines: $MA = \text{load over effort} + \text{lever/pulley}$.	longer effort arm wins.

★ IMPORTANT FOR EXAM

Heat + temperature

⚠ TRAP - read twice

Loss-only + 'energy lost' + convert-miss + ratio-miss

✓ **Try yourself:** Total + convert + ratio + system? **Answer:** Total + MA.

★ HEAT + TEMP SCALES ★

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► Meaning in one line

Energy vs measure , C/F/K , convert.

► Key points

Point	Example
Heat = energy flow + joule + hot to cold.	-
Temp scales + convert + ice/steam points.	0 C = 32 F = 273 K.

★ IMPORTANT FOR EXAM

Till end: transfer + light + sound!

⚠ TRAP - read twice

Same-only + 'heat asked' + unit-miss + convert-miss

✓ **Try yourself:** Energy + degree + convert + points? **Answer:** Heat + C/F/K.